



LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Linear Analysis, MATB24
October 27, 2021
Time: 14.00–19.00

You may freely use formulas from the formula sheet. Refer to relevant theorems from the course when you use them. Write clearly with concise motivations and explain your steps. In each of the six problems a maximum of five points can be achieved.

1. For $n \in \mathbb{N}$ let

$$f_n: [0, \infty) \rightarrow \mathbb{R}, \quad f_n(x) = \frac{2 + nx}{e^x + nx^2}.$$

- Prove that (f_n) converges pointwise on $[0, \infty)$ as $n \rightarrow \infty$ and determine the limit function.
 - Prove that the convergence is uniform on $[1, 2]$, but not on $[0, 1]$.
2. Find a power series solution $u = u(x)$ to the problem

$$4xu'' + 6u' - u = 0, \quad u(0) = 1.$$

Also determine the radius of convergence of the obtained power series and write the solution in closed form for $x > 0$ and $x < 0$.

3. Let $f: \mathbb{R} \rightarrow \mathbb{C}$ be the 2π -periodic function that satisfies

$$f(t) = \begin{cases} (t + \pi)e^{it}, & -\pi < t < 0, \\ te^{it}, & 0 \leq t \leq \pi. \end{cases}$$

- Decide whether $\sum_{n \in \mathbb{Z}} |\hat{f}(n)|$ is finite or infinite without using an explicit computation of the Fourier coefficients of f .
 - Compute the (complex) Fourier series of f .
 - Determine the value of $\sum_{n \in \mathbb{Z}} \hat{f}(n)$ if it exists.
4. Decide whether or not a function with the respective properties exists. If yes, provide an example; if not, prove that no such function can exist.
- a) $f \in L^1(\mathbb{T})$ with

$$\hat{f}(n) = \begin{cases} 27, & n = 20, \\ 10, & n = 21, \\ 0, & n \in \mathbb{Z} \setminus \{20, 21\}. \end{cases}$$

- $f \in L^1(\mathbb{T})$ with $\hat{f}(n) = 1 - \frac{1}{n}$ for all $n \in \mathbb{Z} \setminus \{0\}$.
- $f \in L^2(\mathbb{T})$ with $\hat{f}(n) = in^{-1/2}$ for all $n \in \mathbb{Z} \setminus \{0\}$.

Please, turn over!

5. Find a solution $u = u(x, t)$ for $(x, t) \in [0, \pi/2] \times [0, \infty)$ to the following problem for the heat equation:

$$\begin{cases} u_t = 2u_{xx}, & 0 < x < \pi/2, \ t > 0, \\ u(0, t) = u_x(\pi/2, t) = 0, & t > 0, \\ u(x, 0) = x(\pi - x), & 0 < x < \pi/2. \end{cases}$$

Also briefly discuss the regularity of your solution and in which sense the initial condition is satisfied for it.

6. Let $f, g: \mathbb{R} \rightarrow \mathbb{C}$,

$$f(t) = |t|e^{-|t|}, \quad g(t) = \begin{cases} 0, & t < 0, \\ e^{-2t}, & t \geq 0. \end{cases}$$

- a) Compute the Fourier transforms of f and g .
b) Use the results to determine the integrals

$$\int_{-\infty}^{\infty} \frac{(1-x^2)^2}{(1+x^2)^4} dx \quad \text{and} \quad \int_{-\infty}^{\infty} \frac{1-x^2}{(1+x^2)^2(4+x^2)} dx.$$